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Solving polynomial systems with Gröbner bases

Alessio Caminata

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Introduction

“Polynomial systems are ubiquitous in Cryptography.”

- ▶ Many problems in Cryptography (symmetric or asymmetric) can be modeled as a system of multivariate polynomial equations over a finite field (**Algebraic Attack**).
- ▶ It is important to have efficient methods to solve those systems and estimate their complexity.
- ▶ Gröbner bases provide such a method.

Goal: Explain the connection between Gröbner bases and polynomial systems and how to compute them.

Menu

1. Gröbner bases and polynomial systems
2. Algorithms for computing Gröbner bases
3. Solving degree and related invariants
4. Change of bases

Part I

Gröbner bases and polynomial systems

Term orders

$\mathbb{k}$ field, $R = \mathbb{k}[x_1, \dots, x_n]$ polynomial ring.

A **term order** τ on R is a total order $\leq_\tau$ on the set $\mathbb{T}$ of all terms of R :

1. For all $m_1, m_2, m \in \mathbb{T}$ if $m_1 \leq_\tau m_2$ then $m_1 m \leq_\tau m_2 m$.
2. For all $m_1, m_2 \in \mathbb{T}$ such that $m_1 \mid m_2$ then $m_1 \leq_\tau m_2$.

Example

- Lexicographic order (Lex).

$\mathbf{x}^\alpha \geq_{\text{Lex}} \mathbf{x}^\beta$ if and only if the first non-zero entry from left to right of $\alpha - \beta \in \mathbb{Z}^n$ is positive.

- Degree Reverse Lexicographic (DegRevLex , for short DRL)

$\mathbf{x}^\alpha \geq_{\text{DRL}} \mathbf{x}^\beta$ if and only if $|\alpha| > |\beta|$ or $|\alpha| = |\beta|$ and the first non-zero entry from right to left of $\alpha - \beta \in \mathbb{Z}^n$ is negative.

A term order τ is **degree compatible** if for all $m_1, m_2 \in \mathbb{T}$ such that $\deg(m_1) < \deg(m_2)$ then $m_1 <_\tau m_2$.

Initial ideal

$0 \neq f = \sum_{i \in \mathcal{I}} a_i m_i$, with $a_i \in \mathbb{k} \setminus \{0\}$ and $m_i \in \mathbb{T}$ distinct terms.
The **initial term** or **leading term** of f (w.r.t a term order τ) is

$$\text{in}_\tau(f) = m_j \quad \text{if } m_i < m_j \quad \forall i \in \mathcal{I} \setminus \{j\}.$$

$I \subseteq R$ ideal, the **initial ideal** of I is the (monomial) ideal

$$\text{in}_\tau(I) = (\text{in}_\tau(f) : f \in I \setminus \{0\}).$$

Example $R = \mathbb{k}[x, y], \tau = \text{DRL}$ with $x > y$. Consider $I = (f_1, f_2)$ with

$$\begin{aligned} f_1 &= x^2 - 1, & \text{in}(f_1) &= x^2 \\ f_2 &= xy + x, & \text{in}(f_2) &= xy. \end{aligned}$$

We have $\text{in}(I) \neq (\text{in}(f_1), \text{in}(f_2)) = (x^2, xy)$. Indeed,
 $y + 1 = (-1 - y)f_1 + xf_2 \in I$, thus $\text{in}(y + 1) = y \in \text{in}(I)$.

Gröbner bases

$I \subseteq R$ ideal, τ a term order. A set of polynomials $\mathcal{G}$ of I is a **Gröbner basis** of I (w.r.t τ) if

$$\text{in}_\tau(I) = (\text{in}_\tau(g) : g \in \mathcal{G}).$$

$\mathcal{G}$ is **reduced** if the leading monomials of the elements of $\mathcal{G}$ are monic and

$$m \notin (\text{in}_\tau(h) : h \in \mathcal{G} \setminus \{g\})$$

for any term m appearing in the support of any polynomial $g \in \mathcal{G}$.

Example

We will see that $\mathcal{G} = \{f_1 = x^2 - 1, f_2 = xy + x, f_3 = y + 1\}$ is a DRL-Gröbner basis of $I = (f_1, f_2)$. It is non-reduced, since $xy \in \text{Supp}(f_2)$, but $xy \in (x^2, y) = (\text{in}(f_1), \text{in}(f_3))$.

Theorem

There exists a unique reduced τ -Gröbner basis of I .

Polynomial system solving

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ \vdots \\ f_c(x_1, \dots, x_n) = 0 \end{cases}$$

We associate to the system above the ideal $I = (f_1, \dots, f_c) \subseteq R$.
The **zero locus** of I is the set

$$Z(I) = \{P \in \overline{\mathbb{k}}^n \mid \forall f \in I \ f(P) = 0\}$$

Theorem/Definition

I is **zero-dimensional** if the following (equivalent) facts hold:

1. $|Z(I)| < +\infty$.
2. $\deg(I) := \dim_{\mathbb{k}}(R/I) < +\infty$. (**The degree of I**)
3. For all $i \in \{1, \dots, n\}$, $I \cap \mathbb{k}[x_i] = (h_i(x_i))$ with $h_i \neq 0$.

Polynomial system solving and Gröbner basis

Elimination Theorem (*boosted*)

Let I be a proper and zero-dimensional ideal of R . The reduced lexicographic Gröbner basis of I has the form:

$$\begin{aligned} & p_{n,1}(x_n), \\ & p_{n-1,1}(x_{n-1}, x_n), \dots, p_{n-1,t_{n-1}}(x_{n-1}, x_n), \\ & p_{n-2,1}(x_{n-2}, x_{n-1}, x_n), \dots, p_{n-2,t_{n-2}}(x_{n-2}, x_{n-1}, x_n), \\ & \vdots \\ & p_{1,1}(x_1, \dots, x_n), \dots, p_{1,t_1}(x_1, \dots, x_n), \end{aligned}$$

where $p_{i,t_j} \in R$ for every $i \in \{1, \dots, n\}$, $j \in \{1, \dots, t_i\}$ and $t_1, \dots, t_{n-1} \geq 1$.

Algebraic attack

Let $\mathcal{F} = \{f_1, \dots, f_r\} \subseteq R = \mathbb{F}_q[x_1, \dots, x_n]$ be a polynomial system.
Let $I = (f_1, \dots, f_r)$. In cryptographic applications, often I is zero-dimensional.

A Gröbner basis attack usually proceeds in three steps:

1. Compute a Gröbner basis of I w.r.t a suitable monomial ordering, often DRL.

Algorithms: Buchberger, F_4 , F_5 ,...

2. Extract a univariate polynomial from the system. This is typically done by converting the Gröbner basis to lexicographic order.

Algorithms: FGLM, Gröbner walk,...

3. Find the roots of the univariate polynomial over $\mathbb{F}_q$.

Algorithms: Berlekamp, Cantor-Zassenhaus, ...

Part II

Algorithms for computing Gröbner bases

S-polynomials

- ▶ $f, g \in R = \mathbb{k}[x_1, \dots, x_n]$ of multidegrees $\alpha, \beta \in \mathbb{N}^n$.
- ▶ $\gamma \in \mathbb{N}^n$ such that $\gamma_i = \max\{\alpha_i, \beta_i\}$ for $i = 1, \dots, n$.
 $\mathbf{x}^\gamma = \text{lcm}(\text{in}(f), \text{in}(g))$ is the l.c.m of $\text{in}(f)$ and $\text{in}(g)$.
- ▶ The **S-polynomial** of f and g is defined as

$$S(f, g) = \frac{\mathbf{x}^\gamma}{c_f \text{in}(f)} \cdot f - \frac{\mathbf{x}^\gamma}{c_g \text{in}(g)} \cdot g,$$

$c_f, c_g \in \mathbb{k}$ are the coefficients of $\text{in}(f)$ and $\text{in}(g)$ in f and g .

Example. $f = x^3y^2 - x^2y^3, g = 3x^4y + y^4 \in \mathbb{k}[x, y]$ with DRL $x > y$.

$\text{in}(f) = x^3y^2, c_f = 1, \text{in}(g) = x^4y$, and $c_g = 3$.

The multidegrees are $\deg \text{in}(f) = (3, 2)$ and $\deg \text{in}(g) = (4, 1)$.

Therefore, $\gamma = (4, 2)$ and the S-polynomial is

$$S(f, g) = \frac{x^4y^2}{1 \cdot x^3y^2} \cdot f - \frac{x^4y^2}{3 \cdot x^4y} \cdot g = x \cdot f - \frac{1}{3}y \cdot g = -x^3y^3 - \frac{1}{3}y^5.$$

Division algorithm

Division Algorithm

Given $(f_1, \dots, f_r)$ an ordered list of polynomials of R , then for any $f \in R$ there exist $a_1, \dots, a_r, p \in R$ such that

$$f = a_1 f_1 + \dots + a_r f_r + p$$

with $p = 0$ or a $\mathbb{k}$ -linear combination of monomials none of which divisible by $\text{in}(f_1), \dots, \text{in}(f_r)$. Moreover, if $a_i f_i \neq 0$ then $\text{multideg}(f) \geq \text{multideg}(a_i f_i)$. The polynomial p is called the **remainder** of the division.

Remark

The division (and the remainder) is not unique and may depend on the order of $(f_1, \dots, f_r)$. If the set of polynomials we divide by forms a Gröbner basis, then the order of the elements does not affect the result.

Buchberger's Criterion

Example. $h_1 = xy + 1$, $h_2 = y^2 - 1$ in $R = \mathbb{k}[x, y]$ with t.o. DRL $x > y$.

1. We divide $f = xy^2 - x$ by (h_1, h_2)

$$f = y \cdot h_1 + 0 \cdot h_2 + (-x - y),$$

thus the remainder is $p = -x - y$.

2. We divide $f = xy^2 - x$ by (h_2, h_1)

$$f = x \cdot h_2 + 0 \cdot h_1 + 0,$$

thus the remainder is $p = 0$.

Buchberger's Criterion

Let I be an ideal of R and let $\mathcal{G} = \{g_1, \dots, g_s\}$. Then $\mathcal{G}$ is a τ -Gröbner basis for I if and only if for any i, j ($i \neq j$) the remainder of a division of $S(g_i, g_j)$ by $\mathcal{G}$ is 0.

Algorithm 1 Buchberger's Algorithm

Input: $\mathcal{G} = \{g_1, \dots, g_s\}$ a set of generators of I

Output: A Gröbner basis of I

Pairs $:= \mathcal{G} \times \mathcal{G}$

$m := s$

while Pairs $\neq \emptyset$ **do**

 Choose $(f, g) \in$ Pairs and erase it from Pairs

$g_{m+1} := S(f, g)$

$g_{m+1} := \text{Division}(g_{m+1}, \mathcal{G})$

if $g_{m+1} \neq 0$ **then**

$m := m + 1$

 Pairs $:= \text{Pairs} \cup \{(g_i, g_m) \mid i \in \{1, \dots, m-1\}\}$

$\mathcal{G} := \mathcal{G} \cup \{g_m\}$

end if

end while

return $\mathcal{G}$

Buchberger's Algorithm: an example

$f_1 = x^2 - 1$, $f_2 = xy + x$ in $R = \mathbb{k}[x, y]$ with t.o. DRL with $x > y$.

Set $\mathcal{G} = \{f_1, f_2\}$.

1. We compute the S-polynomial

$$S(f_1, f_2) = \frac{x^2y}{x^2}f_1 - \frac{x^2y}{xy}f_2 = yf_1 - xf_2 = x^2y - y - x^2y - x^2 = -y - x^2,$$

which has initial term $\text{in}(S(f_1, f_2)) = x^2$.

2. We divide $S(f_1, f_2)$ by $\mathcal{G}$

$$-y - x^2 = (x^2 - 1) \cdot (-1) + (xy + x) \cdot 0 + (-y - 1).$$

The remainder is $f_3 = -y - 1 \neq 0$. We add it to the list, which now becomes $\mathcal{G} = \{f_1, f_2, f_3\}$.

3. We have two new pairs to consider: $S(f_1, f_3)$ and $S(f_2, f_3)$. We start by computing

$$S(f_1, f_3) = \frac{x^2y}{x^2}f_1 - \frac{x^2y}{-y}f_3 = yf_1 + x^2f_3 = x^2y - y - x^2y - x^2 = -x^2 - y.$$

We have that $-x^2 - y = yf_1 - xf_2 + 0 \cdot f_3 + 0$. Thus, $S(f_1, f_3)$ has remainder 0 when divided by $\mathcal{G}$. For $S(f_2, f_3)$ we can proceed similarly.

This shows that $\{f_1, f_2, f_3\}$ is a DRL-Gröbner basis of I .

Macaulay matrices

Let $\mathcal{F} = \{f_1, \dots, f_m\} \subseteq R = \mathbb{k}[x_1, \dots, x_n]$ be (non-homogeneous) polynomials and fix a term order τ .

For any degree $d \in \mathbb{Z}_+$ the **Macaulay matrix** $\mathcal{M}_{\leq d}$ of $\mathcal{F}$ has

- ▶ **columns** indexed by terms of R of degree $\leq d$;
- ▶ **rows** indexed by elements $h \cdot f_i$, where h is a term of R such that $\deg(h \cdot f_i) \leq d$.
- ▶ the **entry** (α, β) of $\mathcal{M}_{\leq d}$ is the coefficient of the term in the column β in the polynomial of the row α .

Example

Let $f_1 = x^2 - 1$, $f_2 = xy + x \in R = \mathbb{k}[x, y]$ with term order *DRL* $x > y$.

The Macaulay matrix $\mathcal{M}_{\leq 2}$ of $\mathcal{F} = \{f_1, f_2\}$ is

$\mathcal{M}_{\leq 2}$	x^2	xy	y^2	x	y	1
f_1	1	0	0	0	0	-1
f_2	0	1	0	1	0	0

Linear-algebra-based algorithms

The rows of a Macaulay matrix are elements of the ideal $I = (\mathcal{F})$. Performing row reduction on $\mathcal{M}_{\leq d}$ produces new elements of I and this process is equivalent to computing many S-polynomials simultaneously and reducing them. Thus, one can compute a Gröbner basis by applying Gaussian elimination to the Macaulay matrix $\mathcal{M}_{\leq d}$ for increasing values of d , or for a sufficiently large d .

- *This idea goes back (at least) to Lazard (1983) and it is at the core of many modern algorithms for computing a Gröbner basis such as Faugère's F4 and F5 algorithms as well as the XL family of algorithms.*

Macaulay matrices: an example

Let $f_1 = x^2 - 1$, $f_2 = xy + x \in R = \mathbb{k}[x, y]$ with term order *DRL* $x > y$.

The Macaulay matrix $\mathcal{M}_{\leq 2}$ of $\mathcal{F} = \{f_1, f_2\}$ is

$M_{\leq 2}$	x^2	xy	y^2	x	y	1
f_1	1	0	0	0	0	-1
f_2	0	1	0	1	0	0

Macaulay matrices: an example

Let $f_1 = x^2 - 1$, $f_2 = xy + x \in R = \mathbb{k}[x, y]$ with term order *DRL* $x > y$.

The Macaulay matrix $\mathcal{M}_{\leq 2}$ of $\mathcal{F} = \{f_1, f_2\}$ is

$\mathcal{M}_{\leq 2}$	x^2	xy	y^2	x	y	1
f_1	1	0	0	0	0	-1
f_2	0	1	0	1	0	0

$\mathcal{M}_{\leq 2}$ is already in row reduced echelon form. We pass to $\mathcal{M}_{\leq 3}$.

$\mathcal{M}_{\leq 3}$	x^3	x^2y	xy^2	y^3	x^2	xy	y^2	x	y	1
xf_1	1	0	0	0	0	0	0	-1	0	0
xf_2	0	1	0	0	1	0	0	0	0	0
yf_1	0	1	0	0	0	0	0	0	-1	0
yf_2	0	0	1	0	0	1	0	0	0	0
f_1	0	0	0	0	1	0	0	0	0	-1
f_2	0	0	0	0	0	1	0	1	0	0

Macaulay matrices: an example

Let $f_1 = x^2 - 1$, $f_2 = xy + x \in R = \mathbb{k}[x, y]$ with term order *DRL* $x > y$.

We do Gaussian elimination on $\mathcal{M}_{\leq 3}$ and we keep track of the operations we did.

$\mathcal{M}_{\leq 3}$	x^3	x^2y	xy^2	y^3	x^2	xy	y^2	x	y	1
xf_1	1	0	0	0	0	0	0	-1	0	0
xf_2	0	1	0	0	1	0	0	0	0	0
yf_1	0	1	0	0	0	0	0	0	-1	0
yf_2	0	0	1	0	0	1	0	0	0	0
f_1	0	0	0	0	1	0	0	0	0	-1
f_2	0	0	0	0	0	1	0	1	0	0

Macaulay matrices: an example

Let $f_1 = x^2 - 1$, $f_2 = xy + x \in R = \mathbb{k}[x, y]$ with term order *DRL* $x > y$.

We do Gaussian elimination on $\mathcal{M}_{\leq 3}$ and we keep track of the operations we did.

$\mathcal{M}_{\leq 3}$	x^3	x^2y	xy^2	y^3	x^2	xy	y^2	x	y	1
xf_1	1	0	0	0	0	0	0	-1	0	0
$xf_2 - f_1$	0	1	0	0	0	0	0	0	0	1
$(y-1)f_2$	0	0	1	0	0	0	0	-1	0	0
f_1	0	0	0	0	1	0	0	0	0	-1
f_2	0	0	0	0	0	1	0	1	0	0
$xf_2 - (y+1)f_1$	0	0	0	0	0	0	0	0	1	1

From the last row we obtain the new polynomial $f_3 = y + 1$.
Indeed, $\{f_1, f_2, f_3\}$ is a Gröbner basis.

The mutant trick

Let $\mathcal{F} \subseteq R$ be a finite set of polynomials.

1. Fix an integer $d > 0$, build the Macaulay matrix $\mathcal{M}_{\leq d}$ of $\mathcal{F}$ and perform Gaussian elimination on it to obtain a matrix in reduced row echelon form (RREF).
2. Any row ℓ in the RREF of $\mathcal{M}_{\leq d}$ corresponds to a polynomial h_ℓ of degree $\leq d$. If $\deg(h_\ell) < d$, we add new rows to $\mathcal{M}_{\leq d}$ corresponding to the polynomials uh_ℓ where u is a term, $\deg(uh_\ell) \leq d$ and $uh_\ell \notin \text{rowsp}(\mathcal{M}_{\leq d})$.
3. Repeat the computation of the RREF and the operation of adding new rows, until there are no new rows to add. Denote by $\mathcal{M}_{\leq d}^{\text{mut}}$ the matrix in RREF computed via this algorithm.

Questions and remarks

- **Termination.** It is clear that $\text{rowsp}(\mathcal{M}_{\leq d}^{\text{mut}}) \subseteq (\mathcal{F})_{\leq d}$, where $(\mathcal{F})_{\leq d} = (\mathcal{F}) \cap \{g \in R \mid \deg(g) \leq d\}$. For $d \gg 0$, we have $\text{rowsp}(\mathcal{M}_{\leq d}^{\text{mut}}) = (\mathcal{F})_{\leq d}$ and thus $\text{rowsp}(\mathcal{M}_{\leq d}^{\text{mut}})$ contains a Gröbner basis of $\mathcal{F}$. On the other hand, if no termination criterion is provided, it may be unclear when to stop the previous procedure.
- **Computational cost.** The size of a Macaulay matrix $\mathcal{M}_{\leq d}$ of a polynomial system $\mathcal{F} = \{f_1, \dots, f_r\}$ is $a \times b$ with

$$a = \# \text{rows} = \sum_{i=1}^r \binom{n+d-\deg f_i}{n}, \quad b = \# \text{columns} = \binom{n+d}{d}.$$

In particular, assuming $a \leq b$ the cost of doing row reduction of $\mathcal{M}_{\leq d}$ is bounded from above by $\mathcal{O}(a^2 b)$.

Part III

Solving degree and related invariants

Solving degrees

Let $\mathcal{F}$ be a finite set of polynomials in R .

- ▶ The **solving degree** of $\mathcal{F}$ is the lowest degree d such that the reduced row echelon form (RREF) of $\mathcal{M}_{\leq d}$ contains a Gröbner basis of $\mathcal{F}$, we denote it by $\text{sd}_{\tau}(\mathcal{F})$.
- ▶ The **(mutant) solving degree** of $\mathcal{F}$ is the lowest degree d such that the reduced row echelon form (RREF) of $\mathcal{M}_{\leq d}^{\text{mut}}$ contains a Gröbner basis of $\mathcal{F}$, we denote it by $\text{sd}_{\tau}^{\text{mut}}(\mathcal{F})$.
- ▶ We denote by $\text{max. GB. deg}_{\tau}(\mathcal{F})$ the highest degree of an element in the reduced τ -Gröbner basis of $\mathcal{F}$.

Relations:

$$\text{max. GB. deg}_{\tau}(\mathcal{F}) \leq \text{sd}_{\tau}^{\text{mut}}(\mathcal{F}) \leq \text{sd}_{\tau}(\mathcal{F}).$$

Example. Let $f_1 = x^2 - 1$ and $f_2 = xy + x$ in $R = \mathbb{k}[x, y]$ with the t.o. DRL ($x > y$) as in previous examples. We have that

$$\text{max. GB. deg}_{\text{DRL}}(\mathcal{F}) = 2 < 3 = \text{sd}_{\text{DRL}}^{\text{mut}}(\mathcal{F}) = \text{sd}_{\text{DRL}}(\mathcal{F}).$$

The importance of the solving degree

Let $\mathcal{F}$ be a finite set of polynomials in $R = \mathbb{k}[x_1, \dots, x_n]$.

- The complexity of finding a τ -Gröbner basis with the methods explained before is estimated as

$$\mathcal{O}\left(\binom{n + \text{sd}_{\tau}^{\text{mut}}(\mathcal{F})}{\text{sd}_{\tau}^{\text{mut}}(\mathcal{F})}^{\omega}\right),$$

where $2 < \omega < 3$ denotes the linear algebra constant.

- **Question.** Can we estimate $\text{sd}_{\tau}^{\text{mut}}(\mathcal{F})$ *a priori*, without solving the system?
- **Answer (partial).** We can use related invariants: degree of regularity, first fall degree, last fall degree, Castelnuovo–Mumford regularity,...

Degree of regularity (Bardet, Faugère, Salvy)

$\mathcal{F} = \{f_1, \dots, f_m\}$ finite set of polynomials in $R = \mathbb{k}[x_1, \dots, x_n]$.

$(\mathcal{F}^{\text{top}}) = (f_1^{\text{top}}, \dots, f_m^{\text{top}})$ ideal of homogeneous part of highest degree.

Assume $(\mathcal{F}^{\text{top}})$ is zero-dimensional. The **degree of regularity** of $\mathcal{F}$ is

$$d_{\text{reg}}(\mathcal{F}) = \min\{d \geq 0 \mid (\mathcal{F}^{\text{top}})_d = R_d\}.$$

Example. Let $\mathcal{F} = \{f, g\}$ with $f = x^2 - 1$, $g = y^2 - 1$ in $R = \mathbb{k}[x, y]$.

Then $(\mathcal{F}^{\text{top}}) = (f^{\text{top}}, g^{\text{top}}) = (x^2, y^2)$. We have

$$(x^2, y^2)_2 \neq \langle x^2, xy, y^2 \rangle = R_2$$

$$(x^2, y^2)_3 = \langle x^3, x^2y, xy^2, y^3 \rangle = R_3$$

Therefore, $d_{\text{reg}}(\mathcal{F}) = 3$.

Theorem (Salizzoni)

If $\max \deg f_i \leq d_{\text{reg}}(\mathcal{F})$ and τ is degree compatible. Then

$$\text{sd}_{\tau}^{\text{mut}}(\mathcal{F}) \leq d_{\text{reg}}(\mathcal{F}) + 1.$$

Last fall degree (Huang, Kusters, Yang, Yeo)

$\mathcal{F} = \{f_1, \dots, f_m\}$ finite set of polynomials in $R = \mathbb{k}[x_1, \dots, x_n]$.

The **space of constructible polynomials** $V_{\mathcal{F},i}$ is the smallest $\mathbb{k}$ -vector space such that:

- ▶ $\mathcal{F} \cap R_{\leq i} \subseteq V_{\mathcal{F},i}$,
- ▶ if $f \in V_{\mathcal{F},i}$ and $g \in R_{\leq i - \deg(f)}$, then $fg \in V_{\mathcal{F},i}$.

The **last fall degree** of $\mathcal{F}$ is

$$d_{\mathcal{F}} = \min \{d \in \mathbb{N} \mid f \in V_{\mathcal{F}, \max\{d, \deg(f)\}} \text{ for all } f \in (\mathcal{F})\}.$$

Theorem (C.-Gorla)

If τ is degree compatible, then

$$\text{sd}_{\tau}^{\text{mut}}(\mathcal{F}) = \max \{d_{\mathcal{F}}, \text{max. GB. deg}_{\tau}(\mathcal{F})\}.$$

Last fall degree (Huang, Kusters, Yang, Yeo)

Let $\mathcal{F} = \{f_1, f_2\}$ with $f_1 = x^2 - 1$, $f_2 = xy + x$ in $R = \mathbb{k}[x, y]$.

We know that $(\mathcal{F}) = (x^2 - 1, y + 1)$. Then

- ▶ $V_{\mathcal{F},0} = V_{\mathcal{F},1} = 0$, since $\mathcal{F}$ contains no elements of degree 0 or 1.
- ▶ $V_{\mathcal{F},2} = \langle x^2 - 1, xy + x \rangle$.
Notice that $y + 1 \in (\mathcal{F})_{\leq 2}$, but $y + 1 \notin V_{\mathcal{F},2}$.
- ▶ $V_{\mathcal{F},3} = (x^2 - 1)R_{\leq 1} + (y + 1)R_{\leq 2} = (\mathcal{F})_{\leq 3}$, since

$$y + 1 = xf_2 - (y + 1)f_1 \in V_{\mathcal{F},3}.$$

- ▶ $V_{\mathcal{F},d} = (\mathcal{F})_{\leq d}$ for all $d \geq 3$.

The last fall degree is $d_{\mathcal{F}} = 3$.

Recall that $\text{sd}_{\text{DRL}}(\mathcal{F}) = 3$ and $\max. \text{GB. deg}_{\text{DRL}}(\mathcal{F}) = 2$.

First fall degree (Dubois, Gama / Ding, Schmidt)

$\mathcal{F} = \{f_1, \dots, f_m\}$ finite set of polynomials in $R = \mathbb{F}_q[x_1, \dots, x_n]$.

$\mathcal{S} = \text{Syz}(f_1^{\text{top}}, \dots, f_m^{\text{top}})$ module of **syzygies**.

$\mathcal{T} = \text{Triv}(f_1^{\text{top}}, \dots, f_m^{\text{top}})$ submodule of **trivial syzygies** generated by

$$\{f_i^{\text{top}} e_j - f_j^{\text{top}} e_i : 1 \leq i < j \leq m\} \cup \{(f_i^{\text{top}})^{q-1} e_i : 1 \leq i \leq m\}.$$

The **first fall degree** of $\mathcal{F}$ is

$$d_{\text{ff}}(\mathcal{F}) = \min\{d \in \mathbb{N} : \mathcal{S}_d / \mathcal{T}_d \neq 0\}.$$

Heuristic assumption: $d_{\text{ff}}(\mathcal{F})$ is close to $\text{sd}_{\text{DRL}}(\mathcal{F})$.

Example 1. $q > 3$ odd. $\mathcal{F} = \{x_1 x_2 + x_2, x_2^2 - 1, x_3^{q-1}, x_1^q - x_1\} \subseteq \mathbb{F}_q[x_1, x_2, x_3]$.
Then $d_{\text{ff}}(\mathcal{F}) = 3 < q - 1 = \text{sd}_{\text{DRL}}(\mathcal{F})$.

Example 2. $q > 3$ odd. $\mathcal{F} = \{x_1^2 - 1, x_2^2 - 1\} \subseteq \mathbb{F}_q[x_1, x_2]$.
Then $d_{\text{ff}}(\mathcal{F}) = q > 2 = \text{sd}_{\text{DRL}}(\mathcal{F})$.

Castelnuovo-Mumford regularity

$\mathcal{F} = \{f_1, \dots, f_m\}$ finite set of polynomials in $R = \mathbb{k}[x_1, \dots, x_n]$.

$(\mathcal{F}^h) = (f_1^h, \dots, f_m^h)$ ideal of homogenized polynomials.

The **Castelnuovo-Mumford regularity** of the ideal $I = (\mathcal{F}^h)$ is

$$\text{reg}(\mathcal{F}^h) = \max\{j - i : \beta_{i,j}(I) \neq 0\}.$$

It is a classical invariant in commutative algebra and can be computed from a minimal graded free resolution of I .

Theorem (C.-Gorla)

If $(\mathcal{F}^h)$ is in generic coordinates (*e.g. if $\mathcal{F}$ contains the field equations of $\mathbb{F}_q$*), then

$$\text{sd}_{\text{DRL}}(\mathcal{F}) \leq \text{reg}(\mathcal{F}^h).$$

Part IV

Change of basis

Algebraic attack

Let $\mathcal{F} = \{f_1, \dots, f_r\} \subseteq R = \mathbb{F}_q[x_1, \dots, x_n]$ be a polynomial system.
Let $I = (f_1, \dots, f_r)$. In cryptographic applications, usually I is zero-dimensional.

A Gröbner basis attack usually proceeds in three steps:

1. Compute a Gröbner basis of I w.r.t a suitable monomial ordering, often DRL.

Algorithms: Buchberger, F_4 , F_5 ,...

2. Extract a univariate polynomial from the system. This is typically done by converting the Gröbner basis to lexicographic order.

Algorithms: FGLM, Gröbner walk,...

3. Find the roots of the univariate polynomial over $\mathbb{F}_q$.

Algorithms: Berlekamp, Cantor-Zassenhaus, ...

Staircase and normal form

$I \subseteq R = \mathbb{k}[x_1, \dots, x_n]$ zero-dimensional ideal of degree $D := \dim_{\mathbb{k}}(R/I)$.
 τ term order, G a τ -Gröbner basis of I .

- ▶ The **staircase** of G is a basis of R/I given by

$$\mathcal{W}_{\tau}(G) = \{t \in \mathbb{T} \mid \forall g \in G \quad \text{in}_{\tau}(g) \nmid t\}$$

- ▶ The **border of the staircase** is

$$\mathcal{B}_{\tau}(G) := \{x_i \omega \mid \omega \in \mathcal{W}_{\tau}(G), 1 \leq i \leq n \text{ and } x_i \omega \notin \mathcal{W}_{\tau}(G)\}$$

- ▶ The **normal form** $\text{NF}_{\tau}(f, I)$ of $f \in R$ is the expression of $\bar{f} \in R/I$ with respect to the the basis $\mathcal{W}_{\tau}(G)$.

Example

$$I = (xy^5 - x^2, xy^2 - y) \subseteq \mathbb{F}_{101}[x, y].$$

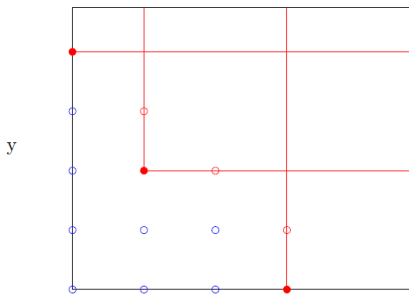
The reduced DRL-Gröbner basis of I is $G = \{xy^2 - y, y^4 - x^2, x^3 - y^3\}$.

The staircase of G is

$$\mathcal{W}_{\text{DRL}}(G) = \{1 < y < x < y^2 < xy < x^2 < y^3 < x^2y\}.$$

The border of the staircase of G is

$$\mathcal{B}_{\text{DRL}}(G) = \{xy^2, x^3, y^4, xy^3, x^2y^2, x^3y\}.$$



FGLM: Idea of the algorithm

I zero-dimensional $\Leftrightarrow \forall i \in \{1, \dots, n\}, I \cap \mathbb{k}[x_i] = (h_i(x_i))$ with $h_i \neq 0$.

Input: G a reduced Gröbner basis of I w.r.t. DRL.

Goal 1: Find $h_i(x_i)$.

Goal 2: Compute the reduced Lex-Gröbner basis of I .

Idea: Since R/I has dimension D , the elements

$$\overline{1}, \overline{x_i}, \overline{x_i^2}, \dots, \overline{x_i^D}$$

are linearly dependent.

Strategy: Compute recursively

$$\text{NF}_{\text{DRL}}(x_i^j, I) = \text{NF}_{\text{DRL}}\left(x_i \cdot \text{NF}_{\text{DRL}}\left(x_i^{j-1}, I\right), I\right) \quad \text{for } j = 1, \dots, D.$$

Express them in the basis $\mathcal{W}_{\text{DRL}}(G)$ until a linear combination

$$\lambda_0 + \lambda_1 x_i + \dots + \lambda_d x_i^d \in I \quad (\text{for } d \leq D)$$

is found.

FGLM Algorithm

Input: $I \neq R$ an ideal, $\mathcal{G}_1$ a reduced Gröbner basis of I w.r.t. $<_1$ and $<_2$ a new term order

Output: $\mathcal{G}_2$ a reduced Gröbner basis of I w.r.t. $<_2$

$L := \text{Sort}([X_1, \dots, X_n], <_2)$ //ordered by $<_2$
 $S := [1]$ //the new staircase under construction for $<_2$
 $V := [1]$ //the NF_1 of S
 $G := []$ //the new G.b.

while $L \neq \emptyset$ **do**

$t := \text{First}(L)$ // $t = \min_{<_2}(L)$

$L := L \setminus [t]$

$v := \text{NF}_1(t)$ // $\text{NF}_1(t) = \text{multiVarDiv}(t, \mathcal{G}_1, <_1)$

$r := \#S$

if $v \in \langle V \rangle_k$ **then**

 find $(\lambda_i)_{i=1}^r$ such that $v = \sum_{i=1}^r \lambda_i V_i$

$G := G \cup \left[t - \sum_{i=1}^r \lambda_i S_i \right]$

else

$S := S \cup [t]$

$V := V \cup [v]$

$L := \text{Sort}(L \cup [X_i t \mid i \in \{1, \dots, n\}], <_2)$

 Remove doubles and multiples of $\text{in}_{<_2}(G)$ from L

end if

end while **return** G

FGLM: Computational cost

- ▶ In FGLM one computes several times $NF(x_k f)$ where f is a polynomial in its normal form. This can be done efficiently using **multiplication matrices** representing the linear map

$$\Phi_k : R/I \longrightarrow R/I, \quad \bar{f} \mapsto \overline{x_k f}$$

with respect to the basis $\mathcal{W}_\tau(G)$.

- ▶ Let G be a reduced Gröbner basis. We have $\text{in}_\tau(G) \subseteq \mathcal{B}_\tau(G)$. In particular, since $|\mathcal{W}_\tau(G)| = D$ and $|\mathcal{B}_\tau(G)| \leq nD$. We obtain the bound $|G| \leq nD$.

Using these observations one can produce an efficient matricial version of FGLM.

Theorem (Faugère, Gianni, Lazard, Mora)

The number of field operations of the matricial FGLM algorithm is bounded from above by

$$\mathcal{O}(nD^3).$$

Gröbner basis attack

Let $\mathcal{F} = \{f_1, \dots, f_r\} \subseteq R = \mathbb{F}_q[x_1, \dots, x_n]$ be a polynomial system.
A Gröbner basis attack usually proceeds in three steps:

1. Compute a Gröbner basis of $I = (\mathcal{F})$ w.r.t DRL.

Complexity estimate:

$$\mathcal{O}\left(\binom{n + \text{sd}_{\text{DRL}}^{\text{mut}}(\mathcal{F})}{\text{sd}_{\text{DRL}}^{\text{mut}}(\mathcal{F})}^\omega\right),$$

where $\text{sd}_{\text{DRL}}^{\text{mut}}(\mathcal{F})$ is the (mutant) solving degree.

2. Extract a univariate polynomial from the system.

Complexity estimate:

$$\mathcal{O}(n(\deg I)^3).$$

3. Find the roots of the univariate polynomial over $\mathbb{F}_q$.

Complexity estimate:

$$\mathcal{O}(\deg I \log(\deg I)(\log(\deg I) + \log(q)) \log \log(\deg I)).$$

Thank you!

Thank you!

Question?

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